# Unit 21 Sample Ratio Mismatch and other Trust-Related.

## (1) Guardrail Metrics 219 ✻ Already discussed.

## (2) Sample Ratio Mismatch 219

### 1) What is Sample Ratio Mismatch (SRM)?

Definition:

→ A SRM occurs when the actual proportion of users assigned to control and treatment groups is significantly different from the intended split.

# Real-world Analogy

⇒ Analogy: Tossing a Fair Coin

You decide to flip a coin 1000 times:

↳ You expect ~ 500 heads and ~ 500 tails (50/50 split)

↳ But you observe: 650 heads, 350 tails.

You suspect something is wrong — maybe the coin is not fair, or you're flipping incorrectly.

Similarly, in experiments:

- If you assign users 50/50 to A/B.
- But end up with 60/40 — SRM is likely → something's broken.

### (2) The Math Behind SRM

→ use a chi-squared goodness-of-fit test to check whether the observed group sizes differ significantly from expected proportions.

Null Hypothesis $H_0$:

| The observed split matches the expected split (no SRM)

Alternative Hypothesis ($H_1$):

The observed split does not match the expected split (SRM present.)

# Formula

Given:

- $n$ = total users
- $p$ = proportions expected for group A
- $O_A$ = observed in group A.
- $O_B = n - O_A$
- $E_A = n \cdot p$; $E_B = n \cdot (1-p)$

Chi-square statistics

$$\chi^2 = \frac{(O_A - E_A)^2}{E_A} + \frac{(O_B - E_B)^2}{E_B}$$

Compare $\chi^2$ to critical value from chi-square distribution with 1 degree of freedom.

(3) Why SRM is Dangerous.

Even if your metric looks fine, SRM is a trust-breaking red flag because:

- It violates randomization, so p-value and CI are no longer valid.
- Any treatment effect you see may be biased.
- Guardrail and primary metrics may be invalid.

## (4) Root Causes of SRM (brief preview)

(Covered in Next ~~Sess~~ Topic "Debugging SRMs 222" —)

- Faulty experiment randomization
- Bug in assignment code (e.g. bot traffic filtered from one group only)
- Timing issues (users join one group more during peak hours)
- Caching or cookie assignment problems
- Incorrect traffic bucketing.

## (5) Summary

| Term | Meaning |
|---|---|
| SRM | Actual group size differs significantly from expected randomization ratio. |
| Why it's bad | Breaks trust in statistical inference (metric no longer reliable). |
| Detect using | Chi-squared test |
| Caused by | Assignment logic bugs, traffic filters, implementation errors. |

## (6) Related / Interlocked Concepts.

1. Guardrail Metrics.
   - → An SRM can make guardrail metrics lie.
   - → If bounce goes up, is it because of treatment or bad assignment Logic?

2. A/A Testing
   - → A/A test (Unit 19) can detect SRM - since metrics should be equal, SRM shows up clearly.

## ③ Debugging SRMs (page 222)

X ① What Does it Mean to Debug SRMs? X

→ When Sample Ratio Mismatch (SRM) is detected, you can't trust the experiment's results. So, debugging SRMs means finding the root cause of the mismatch between expected and actual group sizes.

X ② Common Root Causes (from the book and practical experience).

① Code Bugs in Assignment Logic.

- Your code says 50/50, but somewhere:
  - ↳ Branching is flawed
  - ↳ Assignment cookie is overwritten.
  - ↳ User is re-assigned on reload.

# fix: log assignment logic & double-check bucketing code

② Traffic Filtering or Bot Removal.

- Suppose bot filtering on user eligibility happens after assignment.
- If treatment traffic contains more bots that are filtered out, you see SRM

# Fix: Filter before, assignment or ensure symmetry.

③ Asymmetric Logging Failure

- You assigned users correctly, but didn't log one group properly
  - ↳ logging service was flaky on treatment servers.

⑨ Frontend instrumentation broke for certain browsers.

#Fix : Validate logs at ingestion : compare new traffic vs assigned.

④ Client-side Assignment

- If assignment happens on the client, and:
  - Ad blockers block scripts, or
  - slow connections causes race conditions.

→ users might not be assigned as expected.

#Fix : Move assignment server-side if possible.

⑤ Time-Dependent Assignment

- Suppose you ramp up at different times
- You had a bug during first hour → 90% to control, 10% to treatment.
- Later it's fixed. Final numbers look off.

#Fix : Plot assignment rate over time.

X③ Debugging strategy : step-by-step

| Step | | Action |
|---|---|---|
| 1. | → | Run chi-square test for SRM |
| 2 | → | Break down traffic by dimension (browser, country, platform, etc) |
| 3 | → | Compare assignment logs vs exposure logs. |
| 4 | → | Look at time-series split (users ramp smooth!) |
| 5 | → | Audit assignment code (client/server side!) |

# Relationship between Guardrail Metric and SRM

→ They are deeply interconnected - both serve to ensure the trustworthiness of your online controlled experiments, but they play different roles in your testing life cycle.

# Quick Comparison

| Feature | Guardrail Metric | SRM |
|---|---|---|
| purpose | Detect side effects | Detect broken randomization. |
| Example | Bounce rate, bad ting errors | Imbalance d control / treatment sizes. |
| can affect trust ?. | Yes - if violated, treatment may hurt users | Yes - if SRM exists, all metrics are invalid |
| Link between them | Gaurdrail metrics are meaningless if SRM exists. | Must debug SRM before touching any metrics. |

# Real-world Ancology.

You're testing two diets - Diet A and Diet B.
You assign 50 people to each group randomly.
But due to a bug, Diet A get mostly athletes, and Diet B gets mostly sedentary people.
Now you check a guardrail metric: choles terol levels.

- Diet B has higher cholesterol → you panic!
- But.... wait: you didn't assign people properly
- So your result is biased, and you can't trust

the metric.

> Fix the group imbalance first (SRM) → then trust the cholesterol result (guardrail)

** If SRM exists, your guardrails metrics ** can't be to you.

# Final Rule of Thumb

- ✓ No SRM → Guardrails are meaningful
- ✗ SRM exists → Guardrails are misleading.

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Unit 22 Leakage and Interference between Variants 226

## (1) Examples 227

### (1) What is "Leakage and Interference"?

Before the examples, here's one-line context:

> Leakage happens when users experience more than one variant of the experiment. Interference happens when the treatment to one user affects another user's outcome.

#"Example 227" — Explained in Simple Language.

The authors provide real-world examples where leakage and interference break the independence assumption of A/B testing, which biases results and invalidates metrics. Let's go example by example with analogies, math insight and Python code where helpful.

Example 1 : Shared Computers (Leakage).

# What happen?

- Two user use the same browser/computer (e.g. family laptop)
- One gets assigned to Variant A, another to Variant B
- Assignment is at cookie level, not user level.

# Real-world Analogy.

Imagine you're testing two types of menus in a McDonald's store:

- Some customers see a Red Menu (A)
- Some see a Blue Menu (B)

But the display board changes based on the last customer.

If 3 family members come in and place separate orders and one switches the menu mid-order, each gets a different version based on browser history.

# Again Explaining with another Example.

=> The Core issue:

# Definition:

In many A/B testing systems, especially on websites, users are bucketed into variants (A or B) based on cookies or local storage - i.e. a browser-level identifier.

# Problem:

When multiple people share the same device, their cookie-based identity does not represent a unique user anymore.

This allows variants leakage - where one user see both A and B experiences, either during a sessions or over time.

# Real-world Analogy (MC Donald's Menu)

Let's say you're running an experiment at a MC Donald's store to test two types of digital menu screens:

- Variant A: Red Menu (new layout)
- Variant B: Blue Menu (existing layout).

You want to know:

"Does the red menu increase customer upsells?"

Now imagine:

- A family of 3 comes in.
- They share one kiosk (just like people sharing a laptop browser)
- The first persons see Variant A.
- But then the second person refreshes the browser or logs in/out, and the system reassign them — now they're in Variant B.
- The last person inherits the latest variant, perhaps B again.

The variant assignment is not stable per individual, but volatile based on shared usage.

# Math Perspective: what breaks?

1. Independence assumption of treatments.

In standard A/B testing, we assume:

$$P(Y_i | T_i) \perp P(Y_j | T_j) \quad \text{for } i \neq j$$

This means your behavior (outcome Y) depends only on your assigned treatment T, not others'.

But with shared browsers, this fails.

- User A's experience is influenced by User B's prior treatment (e.g. persistent layout, cookies, recommendations)
- This creates cross-variant contamination.

2. Dilution of Treatment Effect

If 10 users are assigned to Variant A, but 3 of them are "doubtedly see some parts of" Variant B (or vice-versa), then:

$$\hat{Y}_A = \alpha Y_A + (1-\alpha) Y_B.$$

$$\hat{Y}_B = \beta Y_B + (1-\beta) Y_A.$$

where $\alpha, \beta < 1$, due to Leakage from the other variant

This means:

- Estimated means ($\hat{Y}_A$, $\hat{Y}_B$) are now closer to each other than the true effect.
- You'll observe smaller effect sizes → false negatives (Type II errors).

- Example 2: Social Networks (Interference).

# What happens?

In platforms like Facebook, you test a new News Feed ranking algorithm for 50% of users. But users interacts with each other's posts, regardless of variant.

# Real-world Analogy

If you're testing quieter music in half of a bar, and louder music in the other half - but both crowdes share the same dance floor, the effect of one variant spills into the other.

# What's the math here?

Let's say:

- Variant A increases sharing rate $S_A$
- This causes more content availability for everyone - even Variant B users.

So now the outcome $Y_B$ is affected by $S_A$, violating

$$Y_B \not\perp S_A \mid X_B.$$

This is called interference via network effects.

* Known issue: "SUTVA" Violation

SUTVA = Stable Unit Treatment Value Assumption
if one unit's outcome is affected by another's treatment
→ SUTVA is violated → A/B test becomes invalid.

- Example 3: Cross-Device Usage (Leakage)

# What happens?
A user logs into the same service on:
- Phone (assigned Variant A)
- Laptop (assigned Variant B)

Now, both variants influence the same person → treatment effect is confounded.

- Example 4: Search Engine Ranking (Interference)

# What happens?
If different users see different rankings in Google search, their clicks and links influence future rankings.

Over time, control and treatment affect each other's search index.

# The Interference loop.

Let's say
- Variant A increases clicks to Site X.
- Site X becomes more "popular" and appears higher in all variants.
- Now Variant B users see the indirect result of Variant A.

That means

$$P(Y_B / X_B) \text{ is influenced by } X_A$$

This makes A/B separation impossible.

- Example 5: Inventory or Capacity Constraints (Cluster Bias)

# What happens?
You're testing a new layout on e-commerce site that increases conversions (good!).
But your warehouse can't keep up.

So controls users (Variant B) now face out-of-stock products, reducing their performance artificially.

# Real-world Analogy
If you introduces a new fast checkout line (Variant A) and it draws more customers, the regular checkout (Variant B) slows down, causing worse performances than usual.

So you measure:

$$\text{Treatment effect} = \text{Treatment Boost} - \text{Control Penalty}$$

which inflates the observed benefit.

# Differentiate between Leakage Vs Interferences.

| Aspect | Leakage | Interferance |
|---|---|---|
| Definitions | A user see multiple variants (A and B) | A user is outcome is affected by another user's variant. |
| Cause | User/device/session identity is not stable | There is cross-user influence (social, shared resources, etc) |
| Violations | Same user exposed to both treatments | Different users influence each other's outcomes. |
| Comon in | shared devices, multiple logins cross-device usage | social networks, inventory limits, collaborative platforms |
| Problem Type | Contamination via identity leakage | Contamination via cross-unit influence |
| Fix | Assign at stable unit (userID, not cookie/session) | Cluster-based testing, isolation strategies. |
| SUTVA Violated? | ☑ Yes - same unit gets multiple treatments. | ☑ Yes - one unit's outcome depends on another's treatment |

## ② Some Practical Solution 230.

# Goal of This Section:

How do we prevent or mitigate leakage and interference in A/B testing so that treatment effects are unbiased, statistically valid, and interpretable?

# ✓ Practical Solutions (with Simple Explanations & Real-world Analogies)

### 1. Use More Stable Randomization Units.

# What it Means: (Prevents identity-based leakage).

Instead of assigning variants based on cookie, browser, or session, assign based on user ID or account ID.

# Real-world Analogy.

If you're testing a new airline boarding process, don't assign by boarding pass scanner, which can change if people switch lines.

Assign by passenger name - something stable.

# Why it Works

It ensures each user always gets the same variant, regardless of device or session → avoids leakage.

### 2. Cluster Randomization

# What it Means (Avoids cross-user contamination).

Assign entire groups or networks (clusters) to a single variants

Example clusters

- Households
- Offices
- Friends networks
- Schools

# Analogy:
You're testing 2 new school lunch programmes. Instead of assigning individual students randomly, assigns entire classrooms to A or B to avoid interference (e.g. lunch trading).

# Math Insight:
Cluster assignment helps preserve SUTVA

$$Y_i = f(T_{cluster(i)}).$$

where each individual's outcome depends only on their cluster's assignment, not on mixed exposures.

# Interlocks:
This overlaps with:
- SUTVA assumption in causal inference.
- Unit of Analysis ≠ Unit of Randomization (violations cause underpowered tests).

3. Geo-Experiments / Markets-level Randomization.

# What it Means: (~~improve estimation precision~~) (avoid cross-user contamination)
Assign geographical regions (cities, zip codes, countries) to different variants.

# Analogy
Test two type of ad campaigns by showing Campaign A in Kathmandu and Campaign B in Pokhara. Compare city-level revenue.

This avoids interference due to social network effects within regions.

# Use when:
- Users often interact within location (e.g. shopping, delivery radius)
- Can't guarantee user ID consistency.

⚠ Trade-off
- lower statistical power
- Power units to compare → wider CI.

4. Client-Side caching or Sticky Assignments.
# What it Means (Ensure variant consistency).
Once a user is assigned a variant (even by cookie/sessions), store it permanently in their browser or profile.
This avoids flip-flopping between A and B when:
- They open multiple tabs
- They visit from incognito windows.

# Analogy:
Once you're seated in Red section of a stadium, you always get red napkins, even if you wander around.

☑ Benefits
- Reduce leakage
- Doesn't need login-based assignment.

⑤ Holdbacks or Shadow Testing
# What it Means: (Measure algorithm behavior without risk)
Instead of testing full-fledged features impact immediately, silently roll out the features in the background without user interaction.
You collect data without interference.

# Analogy:

You're testing a new recommendation engine, but don't show it to users - just log what it would have recommended, and compare performance.

☑ Benefit

Helps you isolate algorithm behavior before it starts influencing users.

⑥ Event filtering (post-experiment cleanup).

# What it Means (improve estimation precision).

After the experiment ends, filter out corrupted events:

↳ Users who saw both A and B

↳ users with abnormal switches

You exclude their data from final analysis.

⚠ Caution

This introduces bias if the leakage isn't random

# Analogy:

If someone attended both training programs, don't use their test score when comparing which training was better.

## ③ Detecting and Monitoring Interference.

# Goal

> How do you detect and monitor interference when it happens in your A/B tests — especially when it's invisible or silent?

This sections teaches methods to spot violations of SUTVA (Stable Unit Treatment Value Assumption) where a user's outcome is impacted by someone else's treatment.

### ☑ Core Strategies for Detections

1. Vary Treatment Percentage and Observe Non-linear Effects.

# What to do:

Run experiment at different traffic percentage e.g. 1%, 10%, 50%, 90% treatment groups.
Then plot outcome metrics vs treatment fraction.

# Why it works

- If there's no interference, treatment effect should scale linearly with exposure.
- If interference exist (e.g. more treated users "spill" impact), effect may grow non-linearly.

# Real-world analogy:

Imagine spreading perfume in a room
If people are far apart, a small % wearing perfume doesn't affect others.
But at higher % (e.g 50%), the entire room smells like it.

That's how social interference or shared resource interference works

2. Examine Performance of the control Group Over Time.

# what to do

Track control group metrics during the experiments timeline.

# why it works

- In true A/B test, control metrics should be stable over time.
- If interference occurs (e.g. due to resource constraints or viral sharing), control performance might degrade over time as treatment group grows.

# Analogy

Imagine a store introduces express lanes (treatments). As more people use them, the regular lanes (controls) get slower → their satisfaction drops.

3. Use Network Graph Analysis (Social Graphs, shared IPs)

# what to do:

Model users as a network/graph based on:

- Friendship
- Shared household
- Messaging patterns
- IP address.

Then check if treatment assignment is leaking influence to controls nodes.

# Real-world analogy

like second hand smoke - even if you're not smoking (control), your outcomes is affected if you're near someone who is (treatment).

# How it works:

ref:

- $u$ & Control group
- $N(u)$ = neighbors of $u$
- If many $T(v) = 1$ for $v \in N(u)$, and $Y(u)$ improves, that's interference.

4. Look for SRM (Sample Ratio Mismatch).

# Why?

leakage and interference can cause unexpected user inflow into one variant, especially if users switch devices, sessions, or interact with others.

Run SRM checks on:

- Exposures
- logins
- Conversions.

If you see large variance across assignment buckets, it may be due to interference.

This overlaps with:

- Unit 19: A/A tests
- Unit 21: SRM checks.

5. Cross-Device or Multi-Touch Attribution Mismatch

~~If~~ ~~us~~

If a user logs in from phone (assigned to A) and later from laptop (assigned to B), you may see:

- Duplicate conversions
- Blurred attribution

Track:

- Conversion timestamp vs device/variants.
- Reassignments on cross-session collisions.

6 Final Summary Table

| Detection Technique | Detects | What to Look For. |
|---|---|---|
| Treatment fraction vs Metric Curve | Interference | Non-linear trends in metrics. |
| Time Series of Control Metrics | Interference | Degradation or changes during test. |
| Graph-based Neighbor influence | Social or cross-user interference | Treated neighbors influence controls outcome. |
| SRM (Sample Ratio Mismatch) | leakage / bad identity handling. | Bucket imbalance. |
| Cross-device attribution logs. | Identity Leakage | same user treated as both A and B. |
| Regression with neighbor exposure. | statistical interference detection | significant coefficient for ex posure to other treatments. |

# Unit 23 Measuring Long-Term Treatment Effects 235.

## (1) What are long-term effects? (From Unit 23, Page 235).

### (1) Core Idea

In online experiment (like A/B testing), short-term effects are what you observe during the experiment window - often days or weeks.

But the long-term treatment effects are what would happen if the treatment stayed in production permanently.

> Long-term effects = True behavioral impact after people fully adapt.

# Why do short-term ≠ Long-Term?

Because users adapt.

Imagine you redesign a homepage:

- In the first few days, users are confused. Engagement drops.
- But in two months, they've adapted. Now they spend more time, convert more.

So the short-term metric might say "x bad idea" but the long-term effect might be ☑ "very good idea".

# Real-world Analogy: Gym Membership.

Suppose:

- You offer a new gym feature: virtual trainer
- In the first week, people barely use it (change is hard)
- Over 3 months, people start using it daily.

So the short-term metric says "this feature flopped."

But long-term? People stick with your gym longer

This is the long-term treatment effect — what happens after full adoption and habit formation.

## (2) Mathematical Formulas:

Let

- $T(x)$ = Treatment effect at day $x$ (unit: e.g. %, uplift in conversion, revenue/user etc)

- **Short-Term Effect:**
  Measured over the first $k$ days (e.g. 7 days).

$$\text{Short-Term Effect} = \frac{1}{k} \sum_{x=0}^{k-1} T(x)$$

- **Long-Term Effect:**
  Measured over a later time window, say from day $a$ to day $b$:

$$\text{Long-Term Effect} = \frac{1}{b-a+1} \sum_{x=a}^{b} T(x)$$

> In a theoretical sense, if user flow stabilize, we can define:
>
> $$\text{True Long-Term Effect} = \lim_{x \to \infty} T(x).$$

### # Pure Math Example

Let's define a treatment effect that evolves like this:

| Day ($x$) | $T(x)$ = Treatment Effect |
|---|---|
| 0 | -5% |
| 1 | -4% |
| 2 | -2% |
| 3 | -1% |

| | |
|---|---|
| 4 | 0% |
| 5 | +1% |
| 6 | +2% |
| 7 | +3% |
| 8 | +4% |
| 9 | +5% |

Step1: Compute short-term Effect (first 7 days: $n=0$ to $n=6$)

$$\text{Short-term Effect} = \frac{1}{7}\sum_{n=0}^{6} T(n) = \frac{1}{7}(-5-4-3-1+0+1+2)$$

$$= -\frac{9}{7} \approx -1.29\%$$

Interpretation: Initial short-term results look negative

Step2: Compute Long-term Effect (day 3 to day 9: $n=3$ to $n=9$)

$$\text{Long-term Effect} = \frac{1}{7}\sum_{n=3}^{9} T(n) = \frac{1}{7}(-1+0+1+2+3+4+5)$$

$$= \frac{14}{7} = +2.00\%$$

Interpretation: Over time, user behavior adapts, and the treatment becomes beneficial.

# Key Takeaways (Mathematically).

| Concept | Math | Example Outcome |
|---|---|---|
| Short-term avg | Mean of $T(n)$ for early days. | Negative or low if users resist. |
| Long-term avg | Mean over stable period OR limit | Higher if users adapt |
| True long-term | $\lim_{n\to\infty} T(n)$ | Steady state (like 5 above) |

③ Bottom line

| Short - Term | Long - Term |
| --- | --- |
| Captures early reaction | Captures true, stable behavior |
| May be noisy or misleading | often better aligns with business outcomes. |
| Easier to measure | Harder (need s tracking over time). |

(2) Reasons the treatment Effect May Differ Between Short-Term and Long-Term" (page 2.36)

## X ☑ .1. Core Question. X

Why might a treatments (change in product) look good or bad in the short term, but show the opposite behavior in the long run?

## X ☑ .2 Key Reasons for short-term vs long-term Mismatch X

Here are the core 5+ reasons, explained simply, with real-world analogies and backed by maths, behavioral framing and code at the end.

### * A. User learning Curve / Adaption. *

# What's happening?
User need time to learn or adjust to a new feature.

# Example:
- You change a shopping app layout.
- First week: people fumble, cart abandonments rise.
- After 30 days: people use search better → purchase improve.

# Math Frame:

$$T(x) = a(1 - e^{-bx}).$$

A positive function that increases over time (like exponential learning curves).

## * B. Novelty Effect (Initial Curiosity) *

# What is happening?

Users are excited initially, but the excitement fades.

# Examples

- A "spin-the-wheel" gamified coupon.
- Day 1-3: Engagement skyrockets.
- Day 30: users ignore it.

# Math Frame

$$T(x) = q e^{-bx}$$

Exponential decay — spike and drop behavior.

## * C. Network Effects or Peer influence (Interference)

# What is happening?

A user's experience improves only when other also adopt.

# Example:

- You add a "share with friend" feature.
- Initially: no benefit — few users use it.
- Later: more users = more interactions = higher retention.

This relates to interference (Unit 22): treatment effect is non-independent across users.

⚠ Risk:

Short-term results miss the tipping point.

## * D. Behavioral Changes / Habit Formation *

# What's happening?
The treatment induces new habits that pay off over time.

# Example
- Add a daily streak for learning.
- Initially: minor uptick in sessions.
- Long-term: people become daily users.

# Math Frame:

$$T(x) = a \cdot \text{sigmoid}(x - c).$$

where sigmoid models gradual adoption → flattening at a new stable level.

## * E. Delayed Metrics / Downstream Effects *

# What happening?
Some KPIs take time to reflect impact.

# Example:
- You optimize a sign-up funnel → no. short-term revenue increases.
- But 60 days later: higher activation → increased paying users.

# Effect Lag:
short-term:

$$T_{short} = \text{signup uplift}$$

Long-term:

$$T_{long} = \text{revenue uplift via signup} \Rightarrow f(\text{signup}).$$

It's a delayed causal chain.

## 2. Churn or Retention Selectivity.

# What's happening?
A treatment drives bad-fit users to churn, leaving only engaged users

# Example:
- Add a competitive feature in a casual game
- Casual users churn → hardcore users stay and pay more.

# So:
Short-term = Churn spike → looks bad
Long-term = higher monetization from retained cohort.

## 3 ☑ Final Summary Table

| Reason | Short-Term Behavior | Long-Term Behavior |
|---|---|---|
| User Learning | Negative or neutral | - improves over time |
| Novelty spike | High | Wear off. |
| Peer effects | Weak | Gains compound. |
| Habits formation | Slow rise | Stabilizes high. |
| Delayed Metrics | No impact | High impact. |
| Churn filter | look worse | Converts better. |

## ③ Why Measure Long-Term Effect? (Page 238)

### ☑ 1. The Core Reasons.

Short-term metrics can lie.

They may overstate or understate the true value of a treatment because:

- User need time to adapt.
- Habit take time to form.
- Delayed effects (e.g. revenue, retention) don't appear instantly.

> Long-term effect reflect reality
> They answer: "If we ship this to all users permanently, what happens over years/months?"

### ☑ 2. Real-World Analogy

Imagine changing your morning routine

- Day 1: You wake up earlier → tired → worse productivity
- Day 30: Your body adapts → higher energy → better work performance.

If you judged the change on Day 1, you'd cancel it.
But if you measure at Day 30, you'd stick to the improvement.

This is exactly why long-term measurement matters in experiments.

### ☑ 3. Why Measure Long-Term Effects?

# Because short-term signals can mislead

Many treatments appear effective (or harmful) at first, but behave differently over time. Long-term measurement shows true business impact.

# (4) Long-Running Experiments 239.

## ☑ 1. What are Long-Running Experiments?

These are A/B tests that run for weeks or months to measure long-term treatment effects, not just initial reactions

> Their goal: Observe user behavior after adaptation, habit formation, network effects, and delayed outcomes

## ☑ 2. Why Run Long?

- Short-term metrics miss delayed effects.
- Some features look bad early but win long-term (e.g. friction, retention tools).
- Helps estimate steady-state behavior.

## ☑ 3. Challenges of Long-Running Experiments.

| Problem | Why it's a Big Deal |
|---|---|
| Variant exposure | → Users in both A & B groups over time → Interferences. |
| User churn | → Users drop out → survivor bias in long-term metrics. |
| External changes | → New product release or seasonality distort results. |
| Data decay | → Early data becomes less relevant as product evolves. |

## ☑ Real-World Analogy

# Drug trial

- In a week: You see minor side effects.
- In 6 months: You learn if it cures or causes long term damage.

likewise, some A/B test effect take weeks to stabilize.

## ☑ 5. Mathematical Risk Examples Bias Over time.

# Concept

- $U_t$ : Set of active users on day $t$
- $C_t$ : Set of all users who were ever exposed to the treatment by day $t$.
- we're comparing
  - ↳ Observed Effect (biased):

$$\hat{T}_{obs}(t) = \frac{1}{|U_t|} \sum_{i \in U_t} Effect_i$$

  - ↳ True Long-Term Effect (unbiased)

$$T_\infty = \lim_{t \to \infty} \frac{1}{|C_t|} \sum_{i=1}^{|C_t|} Effect_i$$

# Example setup (with numbers).

Suppose:

- New users are added every day
- old users churn quickly
- The treatment effect depends on the type of user,
  - ↳ Early adopters (users who stick around) benefit more.
  - ↳ New or churn-prone users benefit less.

Let's simulate for 5 days.

# Day-by-Day User Data:

| Day | Users who joined | Effect per user | Notes |
|---|---|---|---|
| 0 | User A | +5 | Stays forever |
| 1 | User B, C | +1, +1 | Both churn after day 2 |
| 2 | User D, E | +2, +2 | Both churn after day 3 |
| 3 | User F | +4 | Stays forever |
| 4 | User G | +0 | Churns next day |

Step 1: Cumulative User set by Day 4: $C_4$

All users seen until day 4:

$$C_4 = \{A, B, C, D, E, F, G\}.$$

True cumulative effect:

$$T_{true} = \frac{1}{7}(5+1+1+2+2+4+0) = \frac{15}{7} \approx 2.14$$

Step 2: Active Users on Day 4: $U_4$

Only users still active:

- A (still active)
- F (joined on day 3, still active)

$$U_4 = \{A, F\}$$

Observed effect:

$$T_{obs}(4) = \frac{1}{2}(5+4) = \frac{9}{2} = 4.5$$

# Bias Explained

Because:

- How - affect user (B, C, D, E, F) churned.
- Only high - affect users (A, F) remain active

The observed metric is biased upward.

# Conclusion

if:

- users with higher treatment effects are more likely to stay.
- And you only look at $U_t$, not $C_t$,

Then:

$$\hat{T}_{obs}(t) > \hat{T}_{\infty}$$

This creates a survivor bias - making the treatment seem more successful than it truly is.

## ☑ 7. Design Best Practices for Long-Running Experiment

| Tip | Why it Matters |
|---|---|
| Use consistent cohorts | Avoids churn-based bias. |
| Apply holdouts or re-randomization | Reduces variant interference |
| Monitor for external events | Seasonality / product updates skew results |
| Track adoption curves | To know when behavior stabilizes. |

## ☑ 8. Final Summary.

| | |
|---|---|
| What | Long A/B tests to capture real user behavior |
| Why | Short-term may lie, long-term tells the truth. |
| Challenges | Churn, leakage, seasonality, delta decay. |
| Design tips | Use stable cohorts, monitor interference |
| Risk | Observed effect ≠ true effect due to sampling bias. |

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## (5) Alternative Methods for Long-Running Experiments (page 241)

### 1. Core Idea

When you can't afford to run a long experiment, you still need a way to estimate long-term effects

This section covers alternative methods to predict long-term outcomes without waiting weeks/months.

### 2. Why Alternatives Are Needed.

Running long experiments is often infeasible due to

- Cost (e.g. delayed product rollout = lost revenue)
- Interference over time
- changing user behavior/seasonality
- Business urgency (need fast decisions)

### 3. Alternative Methods (simplified)

- A. Early Predictors or Proxy Metrics

# Idea:

Use short-term metrics that are statistically correlated with a long-term metrics

# Example:

- Click-through rate (CTR) in first 2 days → predicts 30-day engagement.
- Day 1 signup rate → predicts retention at day 30.

# Math:

Let $X$ = early metric, $Y$ = long-term metric

If:

corr $(X, Y)$ is high $\Rightarrow X$ can proxy $Y$

- B. Causal Modeling & ML Estimations.

#Idea:
Build models to estimate the long-term treatment effects based on short-term data + covariates.

#Tools:
- Uplift models
- Causal Forests
- Double ML
- Bayesian structural models.

#Math:

$$Y = f(X, T, \text{User Features}) + \epsilon$$

where $Y$ = long-term KPI, $T$ = treatment indicators, $X$ = early behavior.

- C. Survival Analysis / Retention Modeling.

#Idea:
Use known user life curves (retention or revenue) to project forward.

#Example:
If 7-day retention is 40%, and it decays geometrically, you can extrapolate 30-day retention.

#Math

$$R(t) = R_0 \cdot e^{-bt}$$

#Then compute

$$\text{Expected LTV} = \int_0^{\infty} R(t) \cdot ARPU(t)\, dt$$

- D. Synthetic Control / Re-weighting Techniques.

# Idea:

Use historical control data or matched user cohorts to simulate what would have happened.

Examples:

- CUPED
- PS Weighting
- Matching.

# Math:

Use pre-treatment covariates to re-weight or adjust:

$$\hat{\tau} = \bar{Y}_{T=1} - \bar{Y}_{\text{adjusted } T=0}$$

X ☑ 4. Real-World Analogy X

Suppose you're trying to predict college graduation success. But you only have data from the first semester. Instead of waiting 4 years:

1) You use early GPA as a predictor
2) Or you model graduation probabilities based on study habits, attendance etc.

Same principle applies in A/B tests.

## X 5. Summary Table X

| Method | Use Case | Assumption |
|---|---|---|
| Surrogate Proxies | When strong short-term → long-term correlation exists. | Behavior is stable |
| Causal Models | When enough data to train models exists. | Model captures true effects. |
| Retention Projections | When lifetime curve is predictable | Shape of decay is consistent. |
| Synthetic Controls | When randomization is limited or test can't run long | Covariate balance groups. |

## X 6. Final Takeaways X

- You can't always run experiments for months
- Use smart proxies / models, and statistical tools to forecast long-term impact.
- But always validate these methods whenever possible.